

Model re-entry — operations order (OPORD)

Last synced: 2026-07-30
Owner: Charles + Cursor agent (not a SCOUT handoff)
Status: Executed — archive, 540 surface, Pass B exports on disk.
Charles — personal checkoff: Do not use this OPORD as your to-do list. Use CHARLES_CHECKLIST.md (what you run / prove). This file is ops history and agent phases.

Intent

Re-entry sim work mirrors the document re-entry pattern: park old lab notebooks, build a thin 540_* surface, reuse tier1_* engines. Revolution = new problem approach and entry point; evolution = import battle-tested libraries.
Charles authored the three-step pipeline: assign → score (S_i) → select (top K). Alex endorsed it.
User-facing assignment knob: ρ (rho) = assortativity (0 = max mixing; ρ↑ = sharper match). Legacy τ (temperature) in archived docs = inverse mixing; ρ = 1, σ = 0.65 maps to old τ ≈ 0.65.

Phase 0 — Precondit

#	Step	Owner
0.1	Proceed in this chat (Charles + Cursor agent)	Charles
0.2	Notebook burn: cell index 1 unless skip burn	Agent

Phase 1 — Spec on disk (brief before moves)

#	Step	Deliverable
1.1	Write sports/540_READ_ME_SIM.md	Re-entry sim contract
1.2	Pointer in re_entry/00_READ_ME_FIRST.md	One paragraph
1.3	Update re_entry/02 , 03 : τ → ρ for assignment	Short patches
1.4	Update PARKED_FOR_LATER.md	538D archived; active = 540_*

Gate: Charles approves 540_READ_ME_SIM.md before Phase 2.

Phase 2 — Archive

#	Step	Deliverable
2.1	Create sports/archive/ + README.md	Folder + manifest
2.2	Move into archive	538D_development.ipynb , 538_alex_tier1_model_and_fit.ipynb , 537_Sports_Simulation.ipynb , 535_sports_tier_1.ipynb , 538D_widget_render_test.ipynb , optional 535_*_outputs_backup_*.ipynb
2.3	Grep and fix broken doc pointers	Docs we touch only
2.4	Do not move	530 , tier1_*.py , hero_model_reset_bundle.py , sports_pipeline/

Phase 3 — COMPASS touch (light)

#	Step	Deliverable
3.1	3-Master_Plan/20260727_COMPASS_sim_reentry_status.md	½-page status + claims guard
3.2	Mirror plan if ~/.cursor/plans/ edited	./scripts/mirror_plan.sh (optional)

No mandatory COMPASS agent session.

Phase 4 — SCOUT (reference only)

#	Step	Deliverable
4.1	No SCOUT session required this sprint	—
4.2	Archived notebooks = SCOUT-era lab reference; do not edit	Note in <code>540_READ_ME_SIM.md</code>
4.3	Fresh SCOUT session only if <code>tier1_*</code> bug	Uses 540 brief

Phase 5 — Code (p + Pass B)

#	Step	Deliverable
5.1	p in <code>tier1_pool_assignment.py</code>	<code>exp(-p · (A_i-T_j)²/(2σ²))</code> ; p=0 → uniform among open rosters
5.2	<code>USE_PREFERENTIAL_ATTACHMENT</code> bool in config	α=0 default
5.3	<code>sports/scripts/540_rho_ablation_bundle.py</code>	Low p, high p, sort-and-chop; fix score + winner rule
5.4	Export <code>HEROs_and_PASSes/PASS_B_*</code>	CSVs, PNG, summary, caption
5.5	<code>sports/540_three_step_sim.ipynb</code>	Thin Jupyter orchestration
5.6	Dry-run from repo root	Green run

Pass A (λ knockout): already in `HEROs_and_PASSes/PASS_A_*` — do not redo.

Pass B (p ablation): fix **score** ($S_i / \lambda / L_C$) and **select** (top K); vary assignment only.

Phase 6 — Reconcile Alex bundle

#	Step	Deliverable
6.1	Keep <code>HEROs_and_PASSes/PASS_A_*</code>	Pass A
6.2	<code>PASS_B_README.txt</code>	Cross-link to λ knockout
6.3	Limitation sentence for p pass	caption.txt

Phase 7 — Presentation to Alex (Charles leads)

#	Step	Deliverable
7.1	<code>Model.pdf</code> + 2 min addendum	Three-step pipeline; Pass A + Pass B
7.2	Minimal table	λ knockout vs p ablation arms
7.3	PDFs of updated docs	Charles runs convert scripts locally
7.4	Email / meeting	Figure paths + limitation sentences

Phase 8 — Close-out (when Charles asks)

#	Step
8.1	Git commit (docs + archive + scripts + tracked exports)
8.2	Park stretch in <code>PARKED_FOR_LATER.md</code>

Already done (skip)

- Empirical hero + LPM
- Pass A: λ knockout + `HEROs_and_PASSes/PASS_A_*`
- Re-entry docs 01–03 + `Model.pdf`
- BINDING: `L_net` ≠ advancement; **score** ≠ **select**; unified S_i , $(B-D)=-L_C$

Execution order

1.1 → 1.4 → 2.1 → 2.4 → 3.1 → 5.1 → 5.6 → 6.1 → 6.3 → 7.x → 8.x (optional)
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Assignment formulas (540 contract)

Three-step assignment (generative step 1):

- 1. Draw **A_i** (abilities)
- 2. Draw **T_j** (team target means)
- 3. Soft place: $\pi_{ij} \propto \exp\left(-\rho\left(A_i - T_j\right)^2 / \left(2\sigma^2\right)\right) \times \text{optional } \left(n_j + k\right)^\alpha$

ρ	Meaning
0	Uniform among teams with roster room (max mixing)
1	Moderate ($\equiv$ legacy $\tau=\sigma\approx 0.65$)
High	Sharper match to T_j (not sort-and-chop)

Sort-and-chop: separate benchmark (`method="sort_chop"`) — max assortativity in 537 sense; not $\rho \rightarrow \infty$.

Scoring (step 2): $S_i = A_i - \lambda \cdot L_C$ (λ lives here).

Selection (step 3): winner rule — v1 **top K**; later soft / stochastic. Pass B fixes λ , **L_C**, and K.

Anti-patterns

- Adding cells to archived notebooks
- Duplicating `tier1_*` logic inside 540
- Claiming ρ ablation reproduces hero bin-for-bin
- Renaming ρ back to τ in user-facing prose

See also: `sports/540_READ_ME_SIM.md` , `3-Master_Plan/re_entry/00_READ_ME_FIRST.md`